Unit 2 Homework — Exponent Rules

Complete independently · Exponent Rules

A. Exponent Rules — Product & Quotient

Problem 1
Simplify: $x^5 \cdot x^6$

x^{11}

Problem 2
Simplify: $b^9 \div b^4$

b^5

Problem 3
Simplify: $(4x^2)(2x^5)$

$8x^7$

Problem 4
Simplify: $24m^8 \div 6m^3$

$4m^5$

Problem 5
Simplify: $c^3 \cdot c^4 \cdot c^2$

c^9

Problem 6
Simplify: $(7y^4)(2y)$

$14y^5$

B. Power, Zero & Negative Exponents

Problem 1
Simplify: $(y^5)^2$

y^{10}

Problem 2
Simplify: $(3x^2)^2$

$9x^4$

Problem 3
Simplify: 12^0

1

Problem 4
Simplify: $(8m^3n^2)^0$

1

Problem 5
Simplify: 2^{-3} (write as a fraction)

$\frac{1}{8}$

Problem 6

Simplify: y^{-2} (write as a fraction)

$\frac{1}{y^2}$

Problem 7

Evaluate -2^4 (no parentheses).

-16